PUSH-PULL ALONG A LOCALIZATION AT A SEMI-REPRESENTABLE OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: external tools: Sites hom-sheaf lemma and glueing construction; F(K) coproduct description from the SR definition.

Editorial note. The label lemma-push-pull-localization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Hom sheaf of a representable. Sites, Lemma sites-lemma-hom-sheaf-hU. Second equality: j_{i,*}(F|_{C/U_i}) = SheafHom(h_{U_i}^#, F).
- SR functor F. Hypercoverings, Definition hypercovering-definition-SR-F. F(K) for K a covering family; F(K) = coproduct h_{U_i} as presheaves.

Source reference. Original definition 01G1: sources/stacks-chow/tags.txt, source lines 1890–1890. Prerequisite pointer invoked at proof line(s) 3715; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D7X: sources/stacks-chow/tags.txt, source lines 16572–16572. Prerequisite pointer invoked at proof line(s) 3727; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 04TP: sources/stacks-chow/tags.txt, source lines 6262–6262. Prerequisite pointer invoked at proof line(s) 3732; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a site and K in $SR(\mathcal{C})$. For $\mathcal{F}$ in $Sh(\mathcal{C})$ we have*

$$j_* j^{-1}\mathcal{F} = \mathcal{H}om(F(K)^{\#}, \mathcal{F})$$

where F is as in Hypercoverings, Definition 01G1.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Say $K = \{U_i\}_{i \in I}$. Using the description of the functors j^{-1} and j_* given above we see that

$$j_* j^{-1} \mathcal{F} = \prod\nolimits_{i \in I} j_{i,*}(\mathcal{F}|_{\mathcal{C}/U_i}) = \prod\nolimits_{i \in I} \mathcal{H}om(h^\#_{U_i}, \mathcal{F})$$

The second equality by Sites, Lemma 0D7X. Since $F(K) = \coprod h_{U_i}$ in $PSh(\mathcal{C}$, we have $F(K)^\# = \coprod h^\#_{U_i}$ in $Sh(\mathcal{C})$ and since $\mathcal{H}om(-, \mathcal{F})$ turns coproducts into products (immediate from the construction in Sites, Section 04TP), we conclude. $\square$